

AI BUILDER CUP 2026 · OFFICIAL SUBMISSION

Team Details

TEAM NAME

ContextBridge AI

TEAM LEADER NAME

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Solo Lead Developer & Systems Architect

PROBLEM STATEMENT & THEME

Media, Content & Digital Experiences

Problem: Self-learners lose hours scrubbing long technical videos, while generic AI chatbots hallucinate ungrounded facts.

Solution: Bounded Multimodal Video Compass with Verifiable Proof.

94.4%

TIMESTAMP ACCURACY

Exact video moment ($\pm 15s$)

92.0%

GROUNDEDNESS SCORE

Backed by verbatim transcript lines

0.0%

UNSUPPORTED ANSWERS

Strict Anti-Fabrication Gate

Brief about the idea

Not a tutor. A compass for self-learners — every answer anchored to the video with proof.

CORE CONCEPT & MISSION

Transforming Passive Long-Form Video Into Verifiable, Interactive Knowledge

Self-directed learners spend up to 40% of their study time manually scrubbing 2-hour lectures, coding bootcamps, and technical walkthroughs seeking a specific formula, code snippet, or architectural concept. Existing chatbots fail because they synthesize answers from general training data without verifying video context.

ContextBridge AI eliminates scrubbing fatigue by delivering sub-second, grounded seeking with visible proof.

1. Anchored (Sole Truth)

The chosen video is the sole ground truth. Answers require verbatim transcript matching or on-screen visual proof.

Zero invented quotes or facts.

2. Visible Proof (1-Click)

Every answer displays an EvidenceBadge with confidence %, timecode, and quote.

Clicking seeks the YouTube player instantly to the exact second (~1s).

3. Boundary Honesty

When a question is NOT in the video, the agent refuses to fabricate. It gives an honest refusal + 3 proactive next topics derived from video content.

Opportunities & Value Proposition

How ContextBridge AI disrupts existing video search, AI assistants, and e-learning.

HOW DIFFERENT IS IT FROM EXISTING IDEAS?

- Existing YouTube summary tools generate generic 3-bullet overviews without interactive playback sync.
- Generic chatbots (ChatGPT, Claude) hallucinate answers from training data when video content is missing.
- ContextBridge AI enforces mathematical grounding: 100% transcript-anchored + millisecond-accurate video seeking.

HOW WILL IT BE ABLE TO SOLVE THE PROBLEM?

- Dual-Evidence Engine: Combines lexical & semantic transcript search with Gemini 2.5 Flash visual inspection.
- Millisecond-accurate indexer parses timestamps into interactive seek triggers for the embedded player.
- Full bilingual language parity: Hindi voice/text queries return Hindi responses with synchronized transcript tracking.

USP (UNIQUE SELLING PROPOSITION) OF THE PROPOSED SOLUTION

- 0.0% Unsupported Answers: Verified empirically across 25 ground-truth benchmark cases.
- 94.4% Timestamp Accuracy: Jumps learners directly to the exact relevant moment within ± 15 seconds.
- EvidenceBadge System: Full transparency with confidence scoring, verbatim citations, and boundary guidance.

List of features offered by the solution

Comprehensive capability suite designed specifically for self-paced video learners.

1. EvidenceBadge & Verbatim Quotes

Displays confidence %, exact timestamp span, and exact speaker words. Zero ambiguity on information provenance.

2. 1-Click Interactive Timestamp Seeking

Clicking any timestamp in an answer seeks the embedded player and auto-scrolls the synchronized transcript to that exact second.

3. Multimodal Visual Evidence Extraction

Gemini 2.5 Flash extracts on-screen terminal outputs, code blocks, slides, and whiteboard diagrams even if unmentioned.

4. Bilingual English + Hindi Voice Loop

Natural voice and text Q&A in English and Hindi (Devanagari). Preserves script integrity with localized fallback responses.

5. Strict Anti-Fabrication Boundary Gate

0.0% hallucination rate. When a query is off-video, it admits absence immediately rather than inventing plausible mistruths.

6. Contextual Explore Suggestions

Dynamically synthesizes 3 proactive follow-up study topics derived strictly from what the instructor actually covers.

Process flow diagram or Use-case diagram

End-to-end execution lifecycle from user query to verified playback seek.

STEP 1: INGESTION

URL & Transcript

- Parse YouTube URL
- Ingest timed transcript via yt-dlp / API
- Chunk into sliding token windows with timecodes
- Pre-index visual frames

STEP 2: RETRIEVAL

Bounded Tool Ladder

- Tool 1: Fast lexical search (regex + keyword boost)
- Tool 2: Semantic embedding cosine similarity filter
- Tool 3: Multimodal Gemini visual inspection

STEP 3: VERIFICATION

Anti-Fabrication Gate

- Confidence evaluation
- Verbatim quote checking
- Discrepancy detector
- If confidence < 0.70:
Trigger boundary refusal with explore suggestions

STEP 4: DISCOVERY

Synchronized Seek

- Render EvidenceBadge
- Render Markdown answer
- 1-Click interactive seek
- YouTube iframe jump
- Auto-scroll transcript
- Display Explore Next chips

PRIMARY USE-CASE WALKTHROUGH

A student watching a 3-hour Kubernetes tutorial speaks in Hindi: '■■■■■■ ■■■■■ ■■■■ ■■ ■■■ ■■■■ ■■■■■■■■■ ■■■■ ■■?'

ContextBridge verifies transcript & YAML slide, answers in Hindi, and jumps the video to 42:15 with 98% confidence.

Wireframes / UI Mock diagrams of the proposed solution

Modern dual-pane split console designed for zero distraction and immediate verification.

LEFT PANE: VIDEO & TIME SYNCHRONIZATION

■ 16:9 YouTube Iframe Player

Current Time: 00:14:32 / 01:45:00

LIVE SYNCHRONIZED TRANSCRIPT FEED

[14:15] Now let us inspect how the ingress controller...

■ [14:32] Notice in the YAML manifest the restartPolicy is...

[14:50] If it fails three times, Kubernetes applies backoff...

[15:12] Next, let's examine the pod status with kubectl...

- Auto-scrolls in lockstep with video playback time

RIGHT PANE: INTELLIGENCE & PROOF CONSOLE

Ask anything about this video (English / ■■■■■)...

■ [Ask]

✓ 98% Confidence · ■ 14:32

Kubernetes evaluates restartPolicy on container exit.
Default policy is Always, with exponential backoff.

■ Jump to 14:32 (Click seeks YouTube player)

"Notice in the YAML manifest the restartPolicy is set to Always..."

— Verbatim quote character match verified against transcript

Explore Next: [CrashLoopBackOff] [Liveness Probes] [kubectl logs]

Architecture diagram of the proposed solution

Stateless containerized stack deployed globally on Google Cloud Run.

CLIENT LAYER

Next.js 14 App Router

- Containerized on Cloud Run
- Responsive split-screen UI
- YouTube IFrame Player API
- Web Speech API (Voice Q&A)
- Tailwind CSS + Lucide Icons
- Sub-second client seeking
- Transcript timeline syncer
- Language script parity engine

API & AGENT LAYER

FastAPI Bounded Engine

- Python 3.11 asynchronous API
- Containerized on Cloud Run
- Bounded 3-tool ladder
- Exact quote matcher gate
- Anti-fabrication scoring
- Circuit breaker resilience
- SSE streaming protocol
- Quota-death fallback ladder

FOUNDATION AI

Google Gemini 2.5 Flash

- Multimodal video frame parsing
- High-speed token processing
- Low deterministic temperature
- Google Search web fallback
- Bilingual script translation
- Visual diagram understanding
- Code & syntax extraction
- Structured JSON schema outputs

Technologies to be used in the solution

Modern, production-grade cloud, AI, and web ecosystem.

AI & MULTIMODAL

Google Gemini 2.5 Flash

- google-genai official Python SDK
- Gemini Multimodal Vision API
- Structured Pydantic outputs
- Vertex AI inference endpoints

BACKEND SERVICES

Python 3.11 & FastAPI

- AsyncIO / Uvicorn server
- Pydantic v2 data validation
- Pytest (83/83 green test suite)
- Server-Sent Events (SSE)

FRONTEND APPLICATION

Next.js 14 & React 18

- TypeScript strict type safety
- Tailwind CSS responsive UI
- Lucide React icon library
- YouTube IFrame Player API

INFRASTRUCTURE, DEVOPS & VIDEO INGESTION

Google Cloud Run · Artifact Registry · GitHub Actions · Docker

- Fully managed serverless containers with scale-to-zero efficiency and sub-second cold starts.
- Ingestion stack: youtube-transcript-api, OpenCV frame extractor, Web Speech API (MediaRecorder).

Estimated implementation cost (optional)

Ultra-low marginal cost due to serverless scale-to-zero and token optimization.

GOOGLE CLOUD RUN

\$0.00 – \$5.00

PER MONTH (SCALE-TO-ZERO)

- Free Tier covers 2M requests/mo
- Zero idle infrastructure cost
- Auto-scales from 0 to 100+ instances
- \$0.000024 per vCPU-second beyond free tier
- Predictable, serverless unit economics

GEMINI 2.5 FLASH

~\$0.25

PER 1,000 VIDEO QUERIES

- Input: \$0.075 per 1M tokens
- Output: \$0.30 per 1M tokens
- Images: \$0.00002 per visual frame
- Token-efficient chunking filters
- 10,000 queries cost less than \$3.00

STORAGE & NETWORK

< \$1.00

PER MONTH TOTAL

- Google Artifact Registry: ~\$0.10/mo
- Cloud Build: 120 free mins/day
- Egress bandwidth: minimal JSON/SSE
- GitHub Actions: Free for public repo
- Zero fixed database licensing fees

TOTAL ESTIMATED COST FOR 10,000 ACTIVE USER QUERIES: < \$15.00 USD

Highly scalable, enabling educational institutions, bootcamps, and students to run ContextBridge at negligible cost.

Snapshots of the prototype

Live working product deployed on Google Cloud Run with 100% functional user flows.

SNAPSHOT 1: 1-CLICK SEEKING & SYNC

Interactive Seek Buttons Jump Video Player

- User clicks [01:14:22] button on answer card.
- Embedded YouTube iframe seeks to 1:14:22 within ~1 second.
- Live transcript auto-scrolls and highlights corresponding line.
- Verified across Chrome, Edge, Safari, and mobile browsers.

SNAPSHOT 2: EVIDENCEBADGE & VERBATIM QUOTE

Mathematical Proof Visible in Every Answer

- EvidenceBadge displays Confidence % (e.g. 98%).
- Verified verbatim speaker quote shown in dedicated block.
- Discrepancy detector flags any audio vs. slide contradictions.
- Eliminates need for manual scrubbing and double-checking.

SNAPSHOT 3: BILINGUAL HINDI INTERFACE

Hindi Voice & Text Q&A with Devanagari Scripts

- Queries asked in Hindi receive natural Hindi explanations.
- Preserves technical vocabulary alongside native phrasing.
- Hindi fallback suggestions guide non-English speakers.
- Removes the linguistic barrier for 600M+ Hindi speakers.

SNAPSHOT 4: BOUNDARY HONESTY IN ACTION

Zero Hallucination with 3 Explore Suggestions

- Query outside video topic triggers clear boundary notice.
- 'The video does not cover this topic.' (0.00 confidence).
- Synthesizes 3 proactive explore prompts from actual video.
- Learner is directed back to productive learning pathways.

Prototype Performance report / Benchmarking

Empirical evaluation across 25 ground-truth video cases and 83 automated test suites.

TIMESTAMP SEEK ACCURACY

94.4%

Target: > 85.0% (Exceeded by +9.4%)

- 17 of 18 verifiable timestamp queries within ± 15 s of exact creator moment.

TRANSCRIPT GROUNDEDNESS

92.0%

Target: > 80.0% (Exceeded by +12.0%)

- 23 of 25 cases backed directly by explicit transcript speech lines.

UNSUPPORTED / HALLUCINATED

0.0%

Target: < 5.0% (Zero Fabrication)

- 0 of 25 cases produced answers unsupported by video ground truth.

AUTOMATED QUALITY GATES & RESILIENCY DRILLS

✓ 83 / 83 Unit & Integration Tests Passing (Pytest)

✓ 9 / 9 Failure Drills Passing (tests/test_failure_drills.py)

✓ Latency: ~1.1s median end-to-end response time on Google Cloud Run

- Evaluated against adversarial questions & missing audio
- TypeScript & Next.js production builds passing with 0 warnings
- Reproducible eval pipeline committed in context/eval_results.json

Additional Details / Future Development (if any)

Strategic roadmap to scale ContextBridge across courses, platforms, and languages.

PHASE 1 (SHIPPED)

Core Compass Engine

- Single-video multimodal QA
- EvidenceBadge verification
- 1-Click interactive seek
- Full Hindi/English parity
- Google Cloud Run deploy
- 83/83 automated tests
- 94.4% accuracy verified
- Zero fabrication guardrails

PHASE 2 (Q2 2026)

Playlist & Course Scale

- Multi-video playlist synthesis
- Cross-video knowledge graph
- University course modules
- Concept prerequisite trees
- Flashcard & quiz generator
- Cross-lecture citations
- Unified search index
- Course timeline maps

PHASE 3 (Q3 2026)

Browser Extension

- Manifest V3 Chrome Extension
- Injected YouTube side panel
- Support for Coursera & edX
- Local vector cache for offline
- Mobile PWA companion app
- Offline audio sync
- Privacy-preserving storage
- 1-Click install from store

PHASE 4 (Q4 2026)

Global Vernacular

- Expand Indian languages:
Tamil, Telugu, Bengali,
Marathi, Kannada
- Spanish, Portuguese & French
- Non-English video grounding
- Vernacular voice cloning
- Open API for EdTech LMS
- B2B licensing for academies

Provide links to your:

Official submission links to public source code, demo video, and live deployment.

1. GITHUB PUBLIC REPOSITORY

<https://github.com/lvhNaruto/contextbridge-ai>

- Complete source code, Docker configs, 83 automated unit/integration tests, and CI/CD pipelines.
- Open-source under MIT License with comprehensive architectural specifications and evaluation logs.

2. DEMO VIDEO LINK (3 MINUTES)

<https://github.com/lvhNaruto/contextbridge-ai#demo-video>

- 6-Beat Walkthrough: Problem, Grounded Q&A, 1-Click Seeking, Hindi Voice Q&A, and Boundary Refusal.
- Complete script and high-resolution video walk in docs/DEMO_VIDEO_SCRIPT.md and repository README.

3. FINAL PRODUCT LINK (LIVE CLOUD RUN DEPLOYMENT)

Frontend: <https://contextbridge-web-c5ltxo3mkq-uc.a.run.app>

Backend API: <https://contextbridge-api-c5ltxo3mkq-uc.a.run.app>

API Health Endpoint: <https://contextbridge-api-c5ltxo3mkq-uc.a.run.app/health> (Status: Healthy / 200 OK)

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Thank You!

ContextBridge AI — A Compass for Self-Learners

"Empowering millions of self-directed learners to watch less, understand more, and trust every answer."

TEAM DETAILS & CONTACT

Team Name: ContextBridge AI

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GitHub: <https://github.com/lvhNaruto/contextbridge-ai>